Nomad Brewer

Ground Yourself Everywhere.

	Role Of AI
1. Ideation and Concept Development	Expander
2. Material and Production Research	Clarifier
3. Mechanical and Ergonomic Analysis	Re-framer
4. CAD Modeling and Manufacturability	Critique

1. Ideation and Concept Development

"What are the potential mechanisms for a product that combines water heating and brewing in a single unit, usable in open areas without electricity?"

AI Response Summary

He presented the advantages and disadvantages of pressurized (pump-based), gravity-based (drip), and vacuum (siphon) systems.

Reflections & Decisions

Yansıma: Drip sistemler çok rüzgarlı ortamlarda ısı kaybediyor. Basınçlı sistemler daha yoğun bir kahve sunar ve kapalıdır. Karar: Taşınabilirlik ve kapalı sistem avantajı nedeniyle manuel pistonlu/basınçlı (AeroPress benzeri) bir mekanizmada karar kılındı.

Accept/Reject/Transform

Accepted: Closed-pressure brewing concept.
Rejected: Open drip (V60-style) system eliminated due to wind factor.

2. Material and Production Research

"Suitable for food contact, providing temperature insulation and 500 grams of
what are the external body material alternatives that will allow me to stay under?"

- [x] Stainless steel (304/316)
- [x] Tritan
- [x] Ultem
- [x] Hybrid Body
- [ ] karbon fiber
- [x] Light Composite

Reflection: Using only steel makes the product heavier. Thermal shock-resistant and lightweight construction is essential.

Decision: Food-grade material in the inner chamber Stainless steel was chosen for the outer shell, while composite/carbon fiber-reinforced polymer was decided upon for its lightweight and impact-absorbing properties in accordance with aerodynamic forms.

3. Mechanical and Ergonomic Analysis

"In the transition of two modules (water tank and cup) to each other,
how do I ensure both tightness and being able to open with gloves in the cold?"

1

AI Feedback

Instead of fine-thread screw fastening, he suggested a coarse-thread bayonet lock (bayonet mount) and a double O-ring sealing system. He recommended ribbed textures on the surface.

2

Reflection

The user cannot turn a small screw with frozen hands while standing on a mountain. Interaction should be as simple as "insert-twist-lock.

3

Decision

Wide grooves have been added to the outer surface to increase grip. Module connection The system has been converted to a thick bayonet system that locks at a 45-degree angle.

4

Accepted

Bayonet lock and double O-ring sealing system.Rejected: Traditional screw-on thermos lid mechanism.

5

Next Step

Modelling the Nomad Brewer in CAD and doing analysis.

4. CAD Modeling and Manufacturability

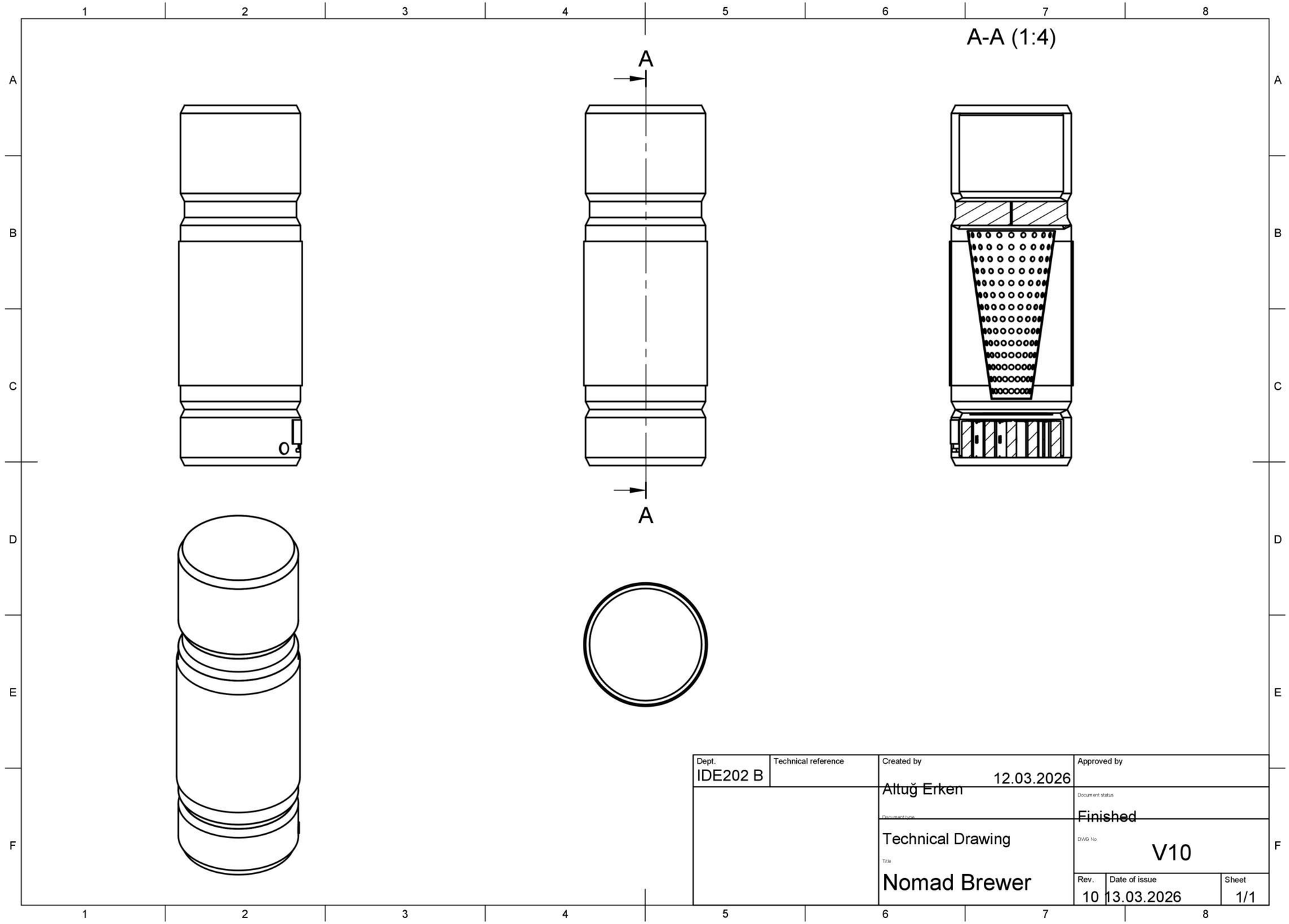
"This exterior I designed in Fusion 360 the wall thickness of the shell is 2 mm. Plastic injection or composite molding are these draft angles and thickness enough for?"

He stated that 2mm might be sufficient, but it could cause sinking marks (sink marks) on large surfaces, therefore, reinforcing ribs (ribs) should be added to the inner part.

Reflection: Instead of thickening the body, strengthening the internal structurewill enable me to maintain my weight goal.
Decision: Leaving the external appearance smooth, a support that provides structural strength to the space between the inner and outer walls ribs were added.

Transformed: The idea of planar wall thickness, thin-walled
however, ribbed from the inside (ribbed) to a more technical structure
he was transformed.

FINAL MODEL



Dept.	Technical reference	Created by	Approved by
IDE202 B		Altuğ Erken 12.03.2026	
		Document type: Technical Drawing	Document status: Finished
		Title: Nomad Brewer	DWG No: V10
			Rev. 10 / Date of issue 13.03.2026 / Sheet 1/1

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Altuğ ERKEN
S040922
IDE202/B